Salary Prediction Using Regression

Harnessing machine learning to optimize compensation strategies and reduce organizational risk through data-driven predictive modeling.

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The High-Stakes Problem

Inefficient compensation models lead to significant financial leakage in human resources.

Financial Leakage

HR firms lose $12M annually due to mis-priced offers based on outdated or anecdotal data.

Operational Risk

Over-paying inflates operational budgets, while under-paying drives a 30% employee turnover rate within the first six months.

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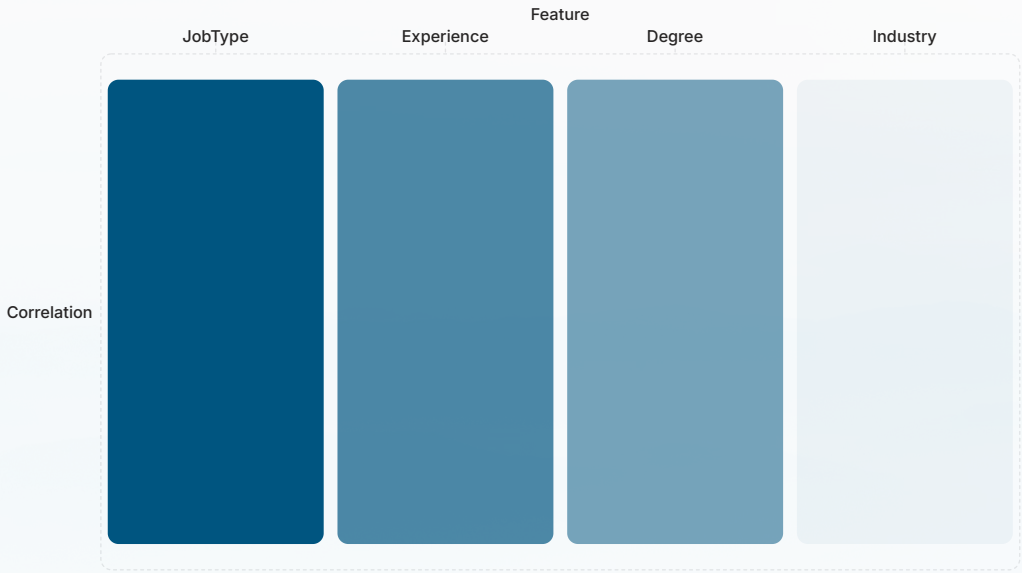

Data at Scale

Leveraging a massive dataset of 1,000,000 job postings to extract deep compensation insights.

- Features: Job type, degree, industry, years of experience, and geographic distance from urban centers.
- Insight: Our analysis identifies **JobType** as the strongest predictor of base salary.

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Baseline Reality Check

Initial industry-standard models provided a foundation, but highlighted the need for more sophisticated regression techniques to lower error margins.

743K

Mean Squared Error

22K

Mean Absolute Error

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Refining Through Feature Engineering

Filter Data

Remove non-correlated CompanyIDs

Encode Groups

Apply mean encodings for job groups

Create Interactions

Add experience-distance terms

Normalize Outliers

Scale and adjust salary distributions

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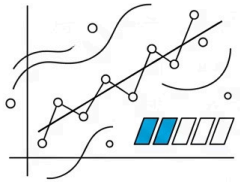
Model Showdown

Comparing various regression architectures to find the optimal balance of precision and interpretability.

89.9%

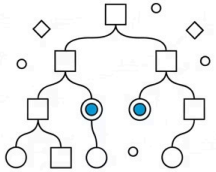
Model Accuracy

Linear Regression



Simple model for linear data.

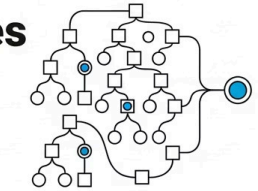
Decision Tree



Uses hierarchical rules.

Random Forest

Aggregates many decision trees.



XGBoost

Sequential boosting often best.

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Production-Ready Pipeline

Seamlessly transitioning from research to production-ready deployment.

Storage

Models and schemata hosted in PostgreSQL.

Automation

Nightly retraining via Airflow DAGs.

Serving

REST API provides real-time predictions.

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Business Impact & ROI

The model delivers tangible financial improvements through accelerated recruitment cycles and talent retention.

22% faster time-to-fill

$

$4.5M annual savings

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Future-Proofing the Model

Staying ahead of global workplace evolution.

New Variables

Integrating remote-work premiums and cost-of-living indices.

Deep Learning

Experimenting with TabNet for richer embeddings.

Drift Monitoring

Real-time alerts if model MAE exceeds 10k.

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Thank You

Thank you for your time and attention today. We hope this overview of our tree-based regression pipeline has provided valuable insights into how data-driven strategies can transform your compensation model. We look forward to the possibility of collaborating on your future initiatives.

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